Work–Energy Theorem & Systems — Worked Solutions

Worksheet convention: use g = 10 m s^{-2}. For numerical work: Equation → Rearrange → Substitute SI values → Final answer with units.

For a single object: $\Delta K = \Sigma W_i$. For the simplified system ledger used in 05a: $\Delta K + \Delta U_g + \Delta U_s = W_{ext}$, with $\Delta E_{mech} = \Delta K + \Delta U_g + \Delta U_s$.

Do not double-count an interaction. If Earth is inside the system, use ΔU_g rather than external gravitational work; similarly for an ideal spring.

Part A · 05a worksheet

1. Lifting a 2.0 kg block

Applied-force work

Equation: W = Fd cosθ

Rearrange: no rearrangement required

Substitute: W = (25)(0.60)cos0°

Answer: $W_{applied}$ = +15 J

Gravity

Equation: W = Fd cosθ

Substitute: W = (20)(0.60)cos180°

Answer: $W_{gravity}$ = -12 J

Change in kinetic energy

Equation: $\Delta K = \Sigma W_i$

Substitute: ΔK = 15 - 12

Answer: ΔK = +3.0 J

Final speed

Equation: K = ½mv²

Rearrange: v = √(2K/m)

Substitute: v = √[2(3.0)/2.0]

Answer: v = 1.73 m s^{-1}.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
System A: block + Earth	+3.0 J	+12 J	0	+15 J	+15 J
System B: block only	+3.0 J	0	0	+3.0 J	+3.0 J

2. Lowering the block

Applied-force work

Equation: W = Fd cosθ

Substitute: W = (15)(0.60)cos180°

Answer: $W_{applied}$ = -9.0 J

Gravity

Equation: W = Fd cosθ

Substitute: W = (20)(0.60)cos0°

Answer: $W_{gravity}$ = +12 J

Change in kinetic energy

Equation: $\Delta K = \Sigma W_i$

Substitute: ΔK = -9 + 12

Answer: ΔK = +3.0 J

Final speed

Equation: K = ½mv²

Rearrange: v = √(2K/m)

Substitute: v = √[2(3.0)/2.0]

Answer: v = 1.73 m s^{-1} downward.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE<sub>mech</s
System A: block + Earth	+3.0 J	-12 J	0	-9.0 J	-9.0 J
System B: block only	+3.0 J	0	0	+3.0 J	+3.0 J

3(a). Projectile: horizontal velocity component

Equation: $v_{0x} = v_0 \cos\theta$

Rearrange: no rearrangement required

Substitute: $v_{0x} = (24)\cos 60°$

Answer: $v_{0x} = 12.0$ m s^{-1}.

3(b). Projectile: stone + Earth system

At the top, the speed is 12.0 m s^{-1}.

Equation: $\Delta K = \frac{1}{2}m(v^2 - v_0^2)$

Substitute: $\Delta K = \frac{1}{2}m(12.0^2 - 24.0^2)$

Answer: $\Delta K = -216m$ J (m in kg).

With no external work:

Equation: $\Delta K + \Delta U_g = W_{ext}$

Rearrange: $\Delta U_g = W_{ext} - \Delta K$

Substitute: $\Delta U_g = 0 - (-216m)$

Answer: $\Delta U_g = +216m$ J.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE<sub>mech</s
System: stone + Earth	-216m J	+216m J	0	0	0

3(c). Projectile: maximum height

Equation: $\Delta K + \Delta U_g = 0$

Use: $\Delta K = \frac{1}{2}m(v^2 - v_0^2)$, and $\Delta U_g = mg\Delta y$

Substitute: $\frac{1}{2}m(12.0^2 - 24.0^2) + m(10)\Delta y = 0$

Rearrange: $\Delta y = [24.0^2 - 12.0^2]/[2(10)]$

Substitute: $\Delta y = (576 - 144)/20$

Answer: maximum height = 21.6 m.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE<sub>mech</s
System: stone + Earth	-216m J	+216m J	0	0	0

3(d). Projectile: stone-only system

Equation: $\Delta K = W_{ext}$

For gravity: $W_{ext} = W_{gravity} = -mg\Delta y$

Substitute: $\frac{1}{2}m(12.0^2 - 24.0^2) = -m(10)\Delta y$

Rearrange: $\Delta y = [24.0^2 - 12.0^2]/[2(10)]$

Answer: maximum height = 21.6 m.

The physics is unchanged; only the bookkeeping changes with the system boundary.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE<sub>mech</s
System: stone only	-216m J	0	0	-216m J	-216m J

4(a)(i). Sliding to a stop

Change in kinetic energy

Equation: $\Delta K = \frac{1}{2}m(v^2 - v_0^2)$

Substitute: $\Delta K = \frac{1}{2}(5.0)(0^2 - 2.0^2)$

Answer: $\Delta K = -10$ J.

Stopping distance

Equation: $W = Fd\cos\theta$

Use: $W_{friction} = \Delta K$

Substitute: $-10 = (16)d\cos180°$

Rearrange: $d = 10/16$

Answer: $d = 0.625$ m.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
System: block	-10 J	0	0	-10 J	-10 J

4(a)(ii). Internal energy generated

Equation: $\Delta U_{int} = f_k d_{rel}$

Substitute: $\Delta U_{int} = (16)(0.625)$

Answer: $\Delta U_{int} = +10$ J.

This is energy generated internally at the contacting surfaces; it is distinct from the work done on either single object.

4(a)(iii). Work done on the stationary belt

Equation: $W = Fd\cos\theta$

Substitute: $W = F(0)\cos\theta$

Answer: work done on the stationary belt = 0 J.

Even though the ideal belt has zero translational displacement, the sliding contact can still generate internal energy in the two surfaces. Real belts also deform slightly.

4(b)(i). Static friction: work on the block

Equation: $W = Fd\cos\theta$

Substitute: $W = (20)(0.50)\cos0°$

Answer: $W_{\text{friction on block}} = +10$ J.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
System: block	+10 J	0	0	+10 J	+10 J

4(b)(ii). Static friction: final speed

Equation: $\Delta K = W_{\text{net}}$

Substitute: $\Delta K = +10$

Answer: $\Delta K = +10$ J.

Equation: $K = \frac{1}{2}mv^2$

Rearrange: $v = \sqrt{(2K/m)}$

Substitute: $v = \sqrt{[2(10)/5.0]}$

Answer: $v = 2.0$ m s^{-1}.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
System: block	+10 J	0	0	+10 J	+10 J

4(b)(iii). Static friction: dissipation

Equation: $\Delta U_{\text{int}} = f_k d_{\text{rel}}$ applies to sliding contact.

Here there is no relative sliding, so $d_{\text{rel}} = 0$.

Answer: sliding-friction dissipation = 0 J.

4(b)(iv). Static friction: work on the belt

Equation: $W = Fd\cos\theta$

Substitute: $W = (20)(0.50)\cos180°$

Answer: $W_{\text{friction on belt}} = -10$ J.

4(c)(i-ii). Kinetic friction can do positive work

(i) Work on the block

Equation: $W = Fd\cos\theta$

Substitute: $W = (0.80)(0.10)\cos0°$

Answer: $W_{\text{friction on block}} = +0.080$ J.

(ii) Final speed

Equation: $\Delta K = W_{net}$

Substitute: $\Delta K = +0.080$

Answer: $K_f = 0.080$ J.

Equation: $K = \frac{1}{2}mv^2$

Rearrange: $v = \sqrt{(2K/m)}$

Substitute: $v = \sqrt{[2(0.080)/5.0]}$

Answer: $v = 0.179$ m s^{-1}.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
System: block	+0.080 J	0	0	+0.080 J	+0.080 J

4(c)(iii-iv). Relative sliding and work on the belt

Acceleration

Equation: $F_{net} = ma$

Rearrange: $a = F_{net}/m$

Substitute: $a = 0.80/5.0$

Answer: $a = 0.160$ m s^{-2}.

Time

Equation: $v = v_0 + at$

Rearrange: $t = (v - v_0)/a$

Substitute: $t = (0.1789 - 0)/0.160$

Answer: $t = 1.118$ s.

Belt displacement

Equation: $d = vt$

Substitute: $d_{belt} = (0.1789)(1.118)$

Answer: $d_{belt} = 0.200$ m.

Relative sliding

Equation: $d_{rel} = d_{belt} - d_{block}$

Substitute: $d_{rel} = 0.200 - 0.100$

Answer: $d_{rel} = 0.100$ m.

(iii) Internal energy generated

Equation: $\Delta U_{int} = f_k d_{rel}$

Substitute: $\Delta U_{int} = (0.80)(0.100)$

Answer: $\Delta U_{int} = +0.080$ J.

(iv) Work on the belt

Equation: $W = Fd\cos\theta$

Substitute: $W = (0.80)(0.200)\cos180°$

Answer: $W_{\text{friction on belt}} = -0.160$ J.

4(c)(v). Energy-flow diagram

The motor replaces the energy removed from the constant-speed belt:

motor / belt

0.160 J→block K

+0.080 J+internal energy

+0.080 J

Check: 0.160 J = 0.080 J + 0.080 J.

Part B · Extra Practice 08a — A to O

The table below gives the requested change-based ledger rather than the original initial/final-energy grid. Values are rounded sensibly; g = 10 m s^{-2}.

Situation	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
A	0	0	0	0	+0 J
B	0	+100 J	0	+100 J	+100 J
C	+20 J	0	0	+20 J	+20 J
D	0	0	0	0	0
E	+15 J	0	0	+15 J	+15 J
F	+5.0 J	+10 J	0	+15.0 J	+15.0 J
G	+5.0 J	0	0	+5.0 J	+5.0 J
H	+550 J	0	0	+550 J	+550 J
I	+550 J	+650 J	0	+1200 J	+1200 J
J	+540 J	0	0	+540 J	+540 J
K	+540 J	+650 J	0	+1190 J	+1190 J
L	+0.200 J	0	0	+0.200 J	+0.200 J
M	+0.200 J	0	-0.200 J	0	0
N	+0.0232 J	+0.0383 J	-0.138 J	-0.0770 J	-0.0770 J
O	+0.0232 J	0	0	+0.0232 J	+0.0232 J

A. Rucksack: system = rucksack only

Equation: $\Delta K = W_{ext}$

Substitute: $0 = W_{ext}$

Answer: $W_{ext} = 0$ J net.

Individually, the lifter does +100 J and gravity does -100 J.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
Energy account	0	0	0	0	+0 J

B. Rucksack: system = rucksack + Earth

Equation: $\Delta U_g = mg\Delta y$

Substitute: $\Delta U_g = (5)(10)(2.0)$

Answer: $\Delta U_g = +100$ J.

Equation: $\Delta K + \Delta U_g = W_{ext}$

Substitute: $0 + 100 = W_{ext}$

Answer: $W_{ext} = +100$ J.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
Energy account	0	+100 J	0	+100 J	+100 J

C. Sandbag, no friction

Equation: $W = Fd\cos\theta$

Substitute: $W = (8)(2.5)\cos0°$

Answer: $W_{ext} = +20$ J.

Equation: $\Delta K = W_{ext}$

Answer: $\Delta K = +20$ J.

Equation: $K = \frac{1}{2}mv^2$

Rearrange: $v = \sqrt{(2K/m)}$

Substitute: $v = \sqrt{[2(20)/10]}$

Answer: $v = 2.0$ m s^{-1}.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
Energy account	+20 J	0	0	+20 J	+20 J

D. Sandbag at constant speed

Pulling work = +20 J. Therefore friction must do -20 J.

Equation: $\Delta K = W_{ext}$

Substitute: $0 = +20 - 20$

Answer: $W_{ext} = 0$ J net.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
Energy account	0	0	0	0	0

E. Sandbag with 2 N friction

Equation: $W_{ext} = W_{pull} + W_{friction}$

Substitute: $W_{ext} = (8)(2.5) + (2)(2.5)\cos180°$

Answer: $W_{ext} = +15$ J.

Equation: $\Delta K = W_{ext}$

Answer: $\Delta K = +15$ J.

Equation: $K = \frac{1}{2}mv^2$

Rearrange: $v = \sqrt{(2K/m)}$

Substitute: $v = \sqrt{[2(15)/10]}$

Answer: $v = 1.73$ m s^{-1}.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
Energy account	+15 J	0	0	+15 J	+15 J

F. Sandbag up shallow incline: bag + Earth

Equation: $\sin\theta = \Delta y/d$

Substitute: $\sin\theta = 0.10/2.5$

Answer: $\theta = 2.29°$.

Equation: $W = Fd\cos\theta$

Substitute: $W_{pull} = (8)(2.5)\cos 2.29°$

Answer: $W_{pull} = +19.98\text{ J} \approx +20.0\text{ J}$.

Equation: $W = Fd\cos\theta$

Substitute: $W_{friction} = (2)(2.5)\cos 180°$

Answer: $W_{friction} = -5.0\text{ J}$.

Equation: $\Delta U_g = mg\Delta y$

Substitute: $\Delta U_g = (10)(10)(0.10)$

Answer: $\Delta U_g = +10\text{ J}$.

Equation: $\Delta K + \Delta U_g = W_{ext}$

Rearrange: $\Delta K = W_{ext} - \Delta U_g$

Substitute: $\Delta K = (19.98 - 5.0) - 10$

Answer: $\Delta K = +4.98\text{ J} \approx +5.0\text{ J}$.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
Energy account	+5.0 J	+10 J	0	+15.0 J	+15.0 J

G. Same incline: sandbag only

Pulling work ≈ +19.98 J; friction work = -5.0 J.

Equation: $W_{gravity} = -mg\Delta y$

Substitute: $W_{gravity} = -(10)(10)(0.10)$

Answer: $W_{gravity} = -10\text{ J}$.

Equation: $\Delta K = W_{ext}$

Substitute: $\Delta K = 19.98 - 5.0 - 10$

Answer: $\Delta K = +4.98\text{ J} \approx +5.0\text{ J}$.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
Energy account	+5.0 J	0	0	+5.0 J	+5.0 J

H. Box up frictionless slope: box only

Equation: $\Delta y = d\sin\theta$

Substitute: $\Delta y = (2.0)\sin 30°$

Answer: $\Delta y = 1.0\text{ m}$.

Equation: $W_{applied} = Fd$

Substitute: $W_{applied} = (600)(2.0)$

Answer: $W_{applied} = +1200\text{ J}$.

Equation: $W_{gravity} = -mg\Delta y$

Substitute: $W_{gravity} = -(65)(10)(1.0)$

Answer: $W_{gravity} = -650\text{ J}$.

Equation: $\Delta K = W_{ext}$

Substitute: $\Delta K = 1200 - 650$

Answer: $\Delta K = +550\text{ J}$.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
Energy account	+550 J	0	0	+550 J	+550 J

I. Same slope: box + Earth

Equation: $\Delta U_g = mg\Delta y$

Substitute: $\Delta U_g = (65)(10)(1.0)$

Answer: $\Delta U_g = +650\text{ J}$.

Equation: $\Delta K + \Delta U_g = W_{ext}$
Substitute: $\Delta K + 650 = 1200$
Rearrange: $\Delta K = 1200 - 650$
Answer: $\Delta K = +550$ J.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
Energy account	+550 J	+650 J	0	+1200 J	+1200 J

J. Rough slope: box only

Applied work = +1200 J; gravity work = -650 J.
Equation: $W_{friction} = Fd\cos180°$
Substitute: $W_{friction} = (5)(2.0)\cos180°$
Answer: $W_{friction} = -10$ J.
Equation: $\Delta K = W_{ext}$
Substitute: $\Delta K = 1200 - 650 - 10$
Answer: $\Delta K = +540$ J.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
Energy account	+540 J	0	0	+540 J	+540 J

K. Rough slope: box + Earth

Equation: $\Delta U_g = mg\Delta y$

Substitute: $\Delta U_g = (65)(10)(1.0)$

Answer: $\Delta U_g = +650$ J.

Equation: $W_{ext} = W_{push} + W_{friction}$

Substitute: $W_{ext} = 1200 - 10$

Answer: $W_{ext} = +1190$ J.

Equation: $\Delta K + \Delta U_g = W_{ext}$

Substitute: $\Delta K + 650 = 1190$

Answer: $\Delta K = +540$ J.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
Energy account	+540 J	+650 J	0	+1190 J	+1190 J

L. Compressed spring: ball only

Convert: 2.5 N cm^{-1} = 250 N m^{-1}.

Equation: $U_s = \frac{1}{2}k(\Delta x)^2$

Substitute: $U_s = \frac{1}{2}(250)(0.040)^2$

Answer: initial spring energy = 0.200 J.

For the ball-only system, this appears as positive external work:

Answer: $W_{ext} = \Delta K = +0.200$ J.

Equation: $K = \frac{1}{2}mv^2$

Rearrange: $v = \sqrt{(2K/m)}$

Substitute: $v = \sqrt{[2(0.200)/0.10]}$

Answer: v = 2.0 m s^{-1}.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
Energy account	+0.200 J	0	0	+0.200 J	+0.200 J

M. Compressed spring: ball + spring

Equation: $\Delta K + \Delta U_s = W_{ext}$

Substitute: +0.200 + (-0.200) = 0

Answer: $W_{ext} = 0$ J; mechanical energy is conserved within the ideal system.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
Energy account	+0.200 J	0	-0.200 J	0	0

N. Pinball: Earth + ball + spring

Convert: m = 0.045 kg; Δy = 0.085 m; d = 0.55 m.

Equation: $\Delta K = \frac{1}{2}m(v^2 - v_0^2)$

Substitute: $\Delta K = \frac{1}{2}(0.045)(1.2^2 - 0.64^2)$

Answer: $\Delta K = +0.0232$ J.

Equation: $\Delta U_g = mg\Delta y$

Substitute: $\Delta U_g = (0.045)(10)(0.085)$

Answer: $\Delta U_g = +0.0383$ J.

Equation: $W_{friction} = Fd\cos180°$

Substitute: $W_{friction} = (0.14)(0.55)\cos180°$

Answer: $W_{ext} = -0.0770$ J.

Equation: $\Delta K + \Delta U_g + \Delta U_s = W_{ext}$

Rearrange: $\Delta U_s = W_{ext} - \Delta K - \Delta U_g$

Substitute: $\Delta U_s = -0.0770 - 0.0232 - 0.0383$

Answer: $\Delta U_s = -0.138$ J.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE<sub>mech</s
Energy account	+0.0232 J	+0.0383 J	-0.138 J	-0.0770 J	-0.0770 J

O. Pinball: ball only

Equation: $\Delta K = W_{ext}$

From N: $\Delta K = +0.0232$ J, so $W_{ext} = +0.0232$ J net.

Individual contributions:

spring work = +0.138 J;

gravity work = -0.0383 J;

friction work = -0.0770 J.

Check: 0.138 - 0.0383 - 0.0770 = 0.0227 J (difference from 0.0232 J is rounding).

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE<sub>mech</s
Energy account	+0.0232 J	0	0	+0.0232 J	+0.0232 J

Part C · Extra Practice 08a — track problem

Track problem 1 — identify the transfers

P1 → P2: gravitational potential energy decreases and kinetic energy increases.

P2 → P3: smooth horizontal motion; K remains constant.

P3 → P4: the blower does positive external work, increasing K.

P4 → P5: friction does negative work, decreasing mechanical energy.

P5 → P6: K decreases to zero while U_s increases.

Track problem 2 — energy account P1 to P6

Gravitational change

Equation: $\Delta U_g = mg\Delta y$

Substitute: $\Delta U_g = (0.120)(10)(-2.5)$

Answer: $\Delta U_g = -3.00$ J.

Blower work

Equation: $W = Fd\cos\theta$

Substitute: $W = (2.2)(1.2)\cos0°$

Answer: $W_{blower} = +2.64$ J.

Friction work

Equation: $W = Fd\cos\theta$

Substitute: $W = (2.6)(1.4)\cos180°$

Answer: $W_{friction} = -3.64$ J.

Therefore $W_{ext} = 2.64 - 3.64 = -1.00$ J.

Equation: $\Delta K + \Delta U_g + \Delta U_s = W_{ext}$

Substitute: $0 - 3.00 + \Delta U_s = -1.00$

Rearrange: $\Delta U_s = -1.00 + 3.00$

Answer: $\Delta U_s = +2.00$ J.

System / scenario	ΔK	ΔU_g	ΔU_s	W_{ext}	ΔE_{mech}
P1 → P6	0	-3.00 J	+2.00 J	-1.00 J	-1.00 J

Track problem 3 — launch speed after decompression

Equation: $\Delta K + \Delta U_s = 0$

Substitute: $K_f - 2.00 = 0$

Answer: $K_f = 2.00$ J.

Equation: $K = \frac{1}{2}mv^2$

Rearrange: $v = \sqrt{(2K/m)}$

Substitute: $v = \sqrt{[2(2.00)/0.120]}$

Answer: $v = 5.77$ m s^{-1}.

Track problem 4 — spring constant

Equation: $U_s = \frac{1}{2}k(\Delta x)^2$

Rearrange: $k = 2U_s/(\Delta x)^2$

Substitute: $k = 2(2.00)/(0.025)^2$

Answer: $k = 6.40 \times 10^3$ N m^{-1}.

Track problem 5 — sign of work

(a) Gravity during A: positive work on the falling ball.

(b) Normal force during B: zero work because it is perpendicular to the displacement.

(c) Blower force during D: positive work.

(d) Friction during E: negative work.

Part D · Quick-check answers

Quick check 1 — system boundary

Use ΔU_g. Gravity is an internal interaction when both the block and Earth are in the system. Do not also include $W_{gravity}$.

Quick check 2 — no double counting

Use ΔU_s and do not also include spring work as external work.

Quick check 3 — friction sign

Yes. A moving conveyor belt can exert kinetic friction in the same direction as a block's displacement, so friction can do positive work on the block.

Quick check 4 — work versus dissipation

Work on an object uses that object's displacement. Sliding dissipation depends on the relative displacement of the two surfaces.

Quick check 5 — mechanical energy

Equation: $\Delta E_{mech} = \Delta K + \Delta U_g + \Delta U_s$

Substitute: $\Delta E_{mech} = 6 - 10 + 1$

Answer: -3 J.

Quick check 6 — object theorem

The change in the object's kinetic energy equals the sum of the work done by all forces on it: $\Delta K = \Sigma W_i$.